HS Data Science 26

Session 01: Orientation, Topics, and Workflow

Prof. Dr. Michael Granitzer

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Seminar Data Science

Focus of This Year's Seminar

- retrieval-augmented generation
- agentic search
- open web retrieval on our own infrastructure

Core Idea

This seminar rewards students who can formulate an interesting question, design a manageable study, and defend their choices clearly.

Speaker notes

Open with the change in seminar logic: fewer predefined topics, stronger emphasis on originality, question quality, and communication.

Seminar Topic

The seminar focuses on retrieval-augmented generation and smart agents in the context of open web search.

Technical Context

- we run our own search engine on the Open Web Index
- students can explore topics through this infrastructure
- the environment supports retrieval experiments and agent workflows

Research Perspective

- search is part of the research object
- retrieval choices can be inspected and compared
- the seminar goes beyond a generic RAG demo

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Speaker notes

Position the seminar technically: shift from generic ML topics to a coherent research environment built on open retrieval and agentic workflows.

Why OWI and Our Search Engine?

The OWI-based environment exists because web retrieval research is otherwise structurally constrained by proprietary search engines and black-box APIs.

What This Enables

- inspect retrieval behaviour
- compare sparse, dense, and hybrid retrieval
- study provenance and ranking choices
- build agentic workflows on transparent search

Consequence for the Seminar

- the search engine is not just a utility
- the retrieval setup itself can be analysed
- good topics exploit the openness of the system

Demo Search

Use the OURRS demo web search for demonstrations and experiments. The API will be available there as well.

How Topic Selection Works

You are expected to generate your own research direction within the seminar theme.

Good Starting Angles

- retrieval quality
- agent planning and verification
- trust and usability
- corpus and query analysis

What Makes a Good Topic

- interesting research question
- manageable scope
- clear evaluation strategy
- relevance to the seminar setup

Example

- weak: *"Build a RAG demo"*
- better: *"When does query reformulation improve agentic search on exploratory tasks?"*

Expected Seminar Work

Scientific Working

- conduct paper research
- identify the gap or tension
- formulate research questions
- document assumptions and limits

Methodology

- implement a prototype or workflow
- run small-scale experiments
- compare alternatives where useful
- explain why the evidence supports the claim

> A modest experiment with a sharp question and clean reasoning is better than an oversized prototype without a defensible result.

Grading

Component	Weight
Independent idea generation	20%
Small-scale but rigorous experiments	15%
Clear methodology and evaluation	15%
Presentation	40%
Report	10%

The asymmetry is intentional: idea generation and presentation are the parts students cannot outsource blindly to AI.

Deliverables

Expected Outputs

- a 1-pager research narrative
- a seminar presentation with a clear research narrative
- a concise report in ACM format
- a Git repository with code, prompts, experiments, or analysis artefacts

Practical Rule

Keep every artefact aligned with one central claim: what question you asked, how you investigated it, and what you learned.

AI Usage

- **Opt-In**: you can use AI for everything, but you have to make it transparent.
- If you use AI, expectations on results, code, report and presentation will be **significantly higher**
- **You fail**
  - if you use AI, but do not make it transparent.
  - if your AI did something wrong (even only a tiny thing) and you did not recognize it.
    - A wrong citation
    - a wrong experiment
    - a wrong claim
    - etc.
  - If your AI did something, and you have
    - no idea what
    - cannot explain concepts, methods, results
    - cannot explain why it did what it did

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Organization

Study Programmes

- BSc Computer Science
- BSc Internet Computing
- MSc Computer Science
- MSc AI Engineering
- open to other faculties

Semester Rhythm

- orientation and topic exploration
- question development
- prototype implementation
- presentations with feedback
- refinement and final report

> Lock the question early, then iterate on evidence and presentation quality.

Tentative Schedule

Date	Topic
Tue 14/04/26	Topic Presentation, Research Hypothesis
Tue 21/04/26	Scientific Working I
Tue 28/04/26	Scientific Working II
Tue 05/05/26	Scientific Working III
Tue 12/05/26	Research Hypothesis Presentation
Tue 19/05 – 23/06	Q&A sessions
Tue 30/06 – 14/07	Student Presentations

All sessions: 10:00–12:00, ITZ R 138.

Next Steps

Before the first week of semester, students should:

- research the field of RAG, agentic search, and web search
- write down one page with a possible topic or research direction
- submit it or present it in the first week of semester

Research-hypothesis presentations are 5–10 minutes.

Spots will be assigned based on the quality of the proposal.

Looking Ahead

Before the next milestone, each student should prepare:

- three candidate topic ideas
- one tentative main research question
- one plausible evaluation strategy
- one short argument for why the question is interesting

> This seminar is about producing an interesting, defensible piece of research in an AI-rich environment where originality, judgment, and communication matter more than routine implementation.